I Did Not Read the README: Beating arXiv:2610.22367 on All Eight Tasks, Under Its Own Grader and a Stricter One

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> “OK.” — the authors′ grader, eight times

Abstract

We report circuits that lower Toffolis × peak qubits (TQ) below the published results on all eight tasks of the arXiv:2610.22367 benchmark. Every result grades OK on the authors′ unmodified grader, and every result also passes a stricter version of that grader that closes a sampling gap we found. Our best secp256k1 point adder reaches 999,283 Toffolis, under one million, on 1,317 qubits, TQ 1,316,055,711: 64.9% below the paper′s 3,746,239,255, with 4.0× fewer Toffolis. It is also below the published point adders of Schrottenloher, IonQ, Luo et al. and the record-and-replay paper, several of which bill more generously than this grader. The ECDSA.Fail paper reports a lower average-billed score, 1.26×10^{9}, which is not comparable to this grader′s per-shot billing. A second point adder, built only from published techniques, reaches TQ 2,255,981,455 (−39.8%) and is released in full so anyone can grade it. The other seven tasks improve by 0.08% to 27.3%. Against the earlier published methods the benchmark itself lists for each task, every result is 5.6× to over 12,000× lower in TQ. The best circuit for each task is released in full, with its grade receipt and an OpenQASM 3 export, and every technique is described. Our compiler and search tools are not released.

1. Introduction

The arXiv:2610.22367 benchmark scores reversible-arithmetic circuits on eight tasks by Toffoli count times peak qubits, with a fixed grader and a stated failure budget per task. Its authors report searched circuits for every task and release the grader, so any result can be re-graded in minutes.

We ask two questions. How far below the published results can each task go? And does the grader's sampling actually certify what each task's contract requires?

On the first, the point adder moves the most: 64.9% below the paper. On the second, we find one task where random sampling accepts streams that are wrong on rare inputs, and we supply reproducers and a patch. Every result is reported under both the stock grader and the stricter one.

The author has no physics degree, no lab and no quantum computer, and worked from a chat window and a C compiler. We mention this not as a flex but because Reviewer 2 was going to ask.

2. Setup

All grades use the authors' grader, andigu/uncompute at commit 64b5a13, unmodified, run as `tools/eval.py <task> --ops <stream> --json`. The C executor shipped with it was used for speed; one stream was re-graded on the pure-Python reference with identical output.

Counting rules, as the grader defines them:

- **Toffolis:** every executed CCX or CCZ on every shot. A CX conditioned on a per-shot classical bit is billed as a Toffoli; a bit-conditioned X is a Clifford.
- **Peak qubits:** the largest qubit id used, plus one. Classical bits are free.
- **Measurement-based uncompute:** an X-basis measurement plus an outcome-conditioned CZ releases a temporary at no Toffoli cost.
- **Shots:** test inputs are drawn from a seed derived from the stream's own hash, so any re-grade of the same file sees the same inputs.

A result counts only if the verdict is OK, failed shots are within the task's limit (for example at most 20 in 100,000 for the point adder), and the stream's sha256 equals the grade's `h_ops`.

3. The point adder

We report two point adders. Both replace the paper's divstep inversion with a different route, and both grade OK on the stock grader.

The published adder. It is built only from published techniques: the ping-pong dialog-GCD division of ECDSA.Fail (arXiv:2609.09582 §5.3.3), its six-stage in-place addition (Algorithm 5), and Gidney's measurement-uncomputed adders (2018); CREDITS.md in the release gives the full lineage. It grades OK at 1,498,991 Toffolis × 1,505 qubits = 2,255,981,455, with 0 failed shots out of 100,000: 39.8% below the paper. We release its full op stream, so anyone can grade it.

Our best adder. It starts from the published adder and adds techniques of our own (Section 4). Its squares use Karatsuba, and every sub-square inside them is an exact square we wrote from the controlled add-subtract and schoolbook description of Litinski (arXiv:2410.00899); each row's carry-in comes free from X gates, since X + Y + 1 = ¬(¬X + ¬Y). One square also takes the high half of its correction from its own input, which removes a copy register. It grades OK at 999,283 Toffolis × 1,317 qubits = 1,316,055,711, under one million Toffolis, with 7 failed shots out of 100,000 against a limit of 20. On a held-out set of 100,000 inputs drawn from an independent seed, it fails on 8, the same 8 inputs on which the earlier variant it was built from fails. We release its full stream as well, so it can be graded the same way. An earlier variant spliced in squaring kernels ported from ECDSA.Fail's published source and graded OK at 1,008,962 Toffolis × 1,317 qubits = 1,328,802,954. That source carries no license, so we neither release that stream nor count it among our results. The squares above were written from Litinski's paper instead, and grade lower.

4. Techniques

An earlier draft of this section was a list of names and a hash. Here are the methods.

This work used 28 techniques, which we name so later work can cite them, and describe each in the table below. The ledger these descriptions come from, as it stood at commit 58948c5 of our private repository, hashes to SHA-256 `0a140ce6b069d0b0212d32ae5dd62fc46ceb2ad2befaf46c2be5f0c91dcf625d`.

A prior-art review of all 28 against primary sources found one direct match: the width bound behind Wishful Thinking (Proven) follows from ECDSA.Fail's Algorithm 5. For the other 27 the underlying principles are published (GF(2) dependency, dirty-ancilla borrowing, wire-lifetime reuse, windowed arithmetic, Booth selection, among others), but the specific constructions were not found in the checked sources. We claim the constructions and their measured results, not the principles.

Where the review found close published work, we credit it. Scarlet's Staircase, Scarlet's Weave and Folding It Wrong (It Worked Anyway) build on ECDSA.Fail's square-specific products and direct accumulation (arXiv:2609.09582, §5.3.4). Gaslighting the Sign Bit builds on Schrottenloher's controlled negation (arXiv:2606.02235, Algorithm 1). Store-Brand Toffolis builds on the quantum Booth multiplier of Álvarez-Sánchez et al. (quant-ph/0611007). Toffolis Without Jurisdiction, Unsupervised Optimism and Checking Their Math, Not Mine build on assertion-based optimization (Häner, Hoefler and Troyer, arXiv:1810.00375). The wire-reuse family builds on Reqomp, Unqomp and dirty-ancilla borrowing. Of those 27, the ones with no closely related published work at all are What Big Parity Doesn't Want You to Know, Plot Armor, Source: Trust Me, I Think This Is Matrix Math, Travel Sized Adders and Whatever Was Lying Around.

#	Technique	What it does
1	Scarlet's Staircase	Builds a square z^2 exactly in 2m bits with m row additions, each carry landing in a bit that is still zero
2	The Kaitlyn Proposal	A bit that equals an XOR of other live bits gets no qubit of its own; it is recovered by CNOTs when needed
3	Scarlet's Weave	Merges the fold steps inside the inversion walk's repeated payload cell
4	Whatever Was Lying Around	Interleaves payload cells, borrowing slots that are still zero right after each walk round
5	Vibes	Sizes each cell's pool of carry qubits to hit a target peak width
6	Unpaid Intern Labor	Lends the idle top wire to every cell
7	What Big Parity Doesn't Want You to Know	Four parity wires per cell come for free from mod-8 invariants, at zero Toffolis
8	Third-Grade Math	Applies the fourth-stage fold of the six-stage addition digit by digit
9	Folding It Wrong (It Worked Anyway)	The merged fold $P{\cdot}F = \Sigma\, s_j{\cdot}low_j + F{\cdot}U$
10	Wishful Thinking (Proven)	Register widths bounded by the walk recurrence, $\lvert t'\rvert \leqslant (\lvert t\rvert + \lvert s\rvert)/2$

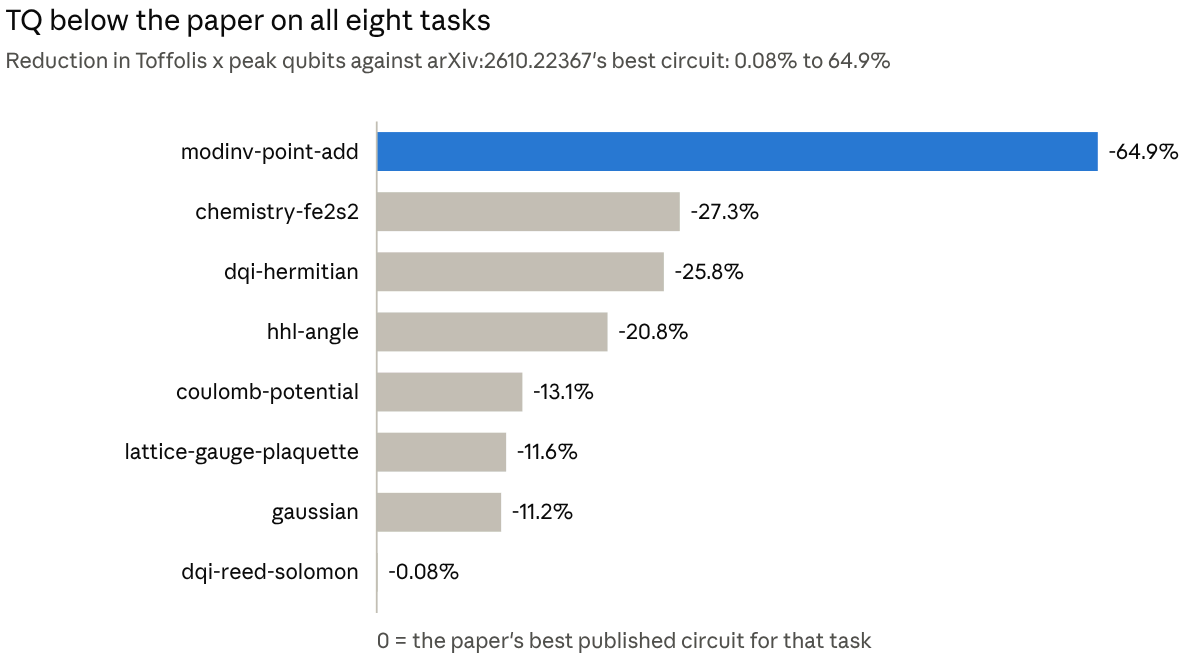
#	Technique	What it does
11	Plot Armor	Window lift: write the constant as $K = 2^n + c$, set bit n with one CNOT, and carry c through a 64-bit window
12	Gaslighting the Sign Bit	Negates a window under a control that is the constant one
13	I Think This Is Matrix Math	Clears a wraparound flag using a comparator on the top 32 bits only
14	Source: Trust Me	A virtual operand: the window's operand is never materialised, it is folded straight into the carries
15	Travel Sized Adders	Wide additions done in per-cell chunks, with the boundary carries erased by top-32 comparators
16	Copy-Pasting to Victory	Applies 13 and 15 to the thirteen adders of the fourth stage
17	Social Anxiety	Fitted carry pools for the walk rounds that sit above the peak target
18	Borrowing Without Asking	Zero extensions and idle tails lent to the fourth-stage adders
19	Strategic Incompetence	Trades part of the failure budget for shorter comparators
20	The Thing I Built for Nothing	A schedule-margin trade; built and graded, but not in the best stream
21	Toffolis Without Jurisdiction	An exact affine census of the compiled stream: finds and deletes gates whose effect is already a known linear function
22	Musical Chairs	Re-homes wire intervals so the Karatsuba squares fit inside the peak
23	Unsupervised Optimism	Clears, at the peak, ancillas that are exact XORs of live wires

#	Technique	What it does
24	Checking Their Math, Not Mine	The affine census of 21, extended across measurement-based uncomputation
25	Squatter's Rights	Renames wire intervals into unused gaps, with every rename checked by hashing the result
26	Store-Brand Toffolis	A 3-Toffoli form of the Booth selection rows in the Karatsuba multiplier, leaving the fix-up terms untouched
27	Close-Enough-for-Government-Work Rounding	Drops low-order product groups and repairs each term inside the task's allowed error band
28	Turning It Off and On Again	Measures out wires that sit idle and rebuilds them when they are next needed

Suggested usage: "We then used Travel Sized Adders, followed by Turning It Off and On Again." No technique changes a task's input or output contract, and each was checked on its sub-block before grading. Not all are exact: some, including Strategic Incompetence (19) and Close-Enough-for-Government-Work Rounding (27), deliberately spend part of a task's failure budget or error band. Every result grades OK under the unchanged task limits.

5. Results

All eight tasks grade OK below the paper's best published circuit, by 0.08% to 64.9% in TQ.



Official grades, andigu/uncompute @64b5a13, stock grader; paper values from each task's README

Table 1. Counts behind each result, ordered as in the chart. Every row is an official OK grade whose stream hash matches its receipt.

Task	Toffolis	Qubits	TQ	Paper TQ	Change	Failed shots (limit)	Construction
modinv-point-add, best	999,283	1,317	1,316,055,711	3,746,239,255	−64.9%	7 (20)	independent
modinv-point-add, published techniques	1,498,991	1,505	2,255,981,455	3,746,239,255	−39.8%	0 (20)	independent, published techniques only
chemistry-fe2s2	9,467	145	1,372,715	1,887,228	−27.3%	0 (0)	edit
dqi-hermitian	707,278	2,328	1,646,543,184	2,219,954,240	−25.8%	59 (100)	edit
hhl-angle	1,087	113	122,831	155,020	−20.8%	0 (0)	edit
coulomb-potential	1,012	139	140,668	161,880	−13.1%	0 (0)	edit
lattice-gauge-plaquette	152,832	1,850	282,739,200	319,985,952	−11.6%	0 (0)	edit
gaussian	1,416	138	195,408	220,037	−11.2%	0 (0)	edit

Task	Toffolis	Qubits	TQ	Paper TQ	Change	Failed shots (limit)	Construction
dqi-reed-solomon	442,043	1,837	812,032,991	812,683,289	−0.08%	8 (100)	edit

TQ charges for every qubit on every Toffoli, so width is already in the score. The point adder uses 1,317 qubits to the paper's 935 and still costs 64.9% less; chemistry uses 145 to 126 and still costs 27.3% less. The dqi-reed-solomon gain is small, 0.08%.

5.1 Against the earlier published methods

Each task's README in the grader lists a baseline built from an earlier published method. Every one of our results beats its baseline, by 5.6× to 12,288×.

Task	Earlier method (as the grader lists it)	Toffolis	Qubits	TQ	Ours, TQ	Ours is lower by
dqi-hermitian	Sakata / Feng–Rao decoder	104,263,342	194,053	2.02×10^{13}	1,646,543,184	12,288×
gaussian	Häner, Roetteler & Svore, piecewise evaluation	23,784	362	8,609,808	195,408	44.1×
modinv-point-add	Luo et al., packed Euclidean walk	69,529,757	838	58,265,936,366	1,316,055,711	44.3×
hhl-angle	Häner, Roetteler & Svore, piecewise evaluation	16,468	300	4,940,400	122,831	40.2×
dqi-reed-solomon	Khattar et al., extended Euclid	5,125,070	1,938	9,932,385,660	812,032,991	12.2×
coulomb-potential	Häner, Roetteler & Svore, piecewise evaluation	6,355	252	1,601,460	140,668	11.4×
lattice-gauge-plaquette	Osorio et al., composed	1,271,424	1,927	2,450,034,048	282,739,200	8.7×
chemistry-fe2s2	Babbush et al., composed	55,473	139	7,710,747	1,372,715	5.6×

These baselines are the benchmark authors' implementations of each earlier method, as listed in the grader at commit 64b5a13. For seven tasks, most of each gap was already closed by the benchmark authors' own circuits, which our results edit; our share there is

the margin in Table 1. The point adder is our own construction: the published-techniques adder alone is 25.8× below its baseline, and our techniques take that to 44.3×.

6. Stricter evaluation

On hhl-angle, the stock grader tests a random sample of accumulator values k, and that sample misses rare wrong inputs. To show it, we broke the authors' own discovered circuit on purpose, two ways, one edit each. The stock grader accepts both. They are in reproducers/ in the release, and one command shows it: decompress reproducers/hhl-angle/coverage-gap-blip/stream.ops.xz and run tools/eval.py hhl-angle on it, which prints verdict OK.

- **coverage-gap-carry30.** One product row's addition into k stops at bit 30, so its carry never reaches the top three bits. Graded OK at 1,345 Toffolis with 0 of 8,747 shots failed, but wrong on 1,221 tested (X, k) lanes, each off by about 2^{30}.
- **coverage-gap-blip.** One CX is removed: the same row's carry into the top bit of k. Graded OK at 1,348 Toffolis with 0 of 8,748 shots failed, but wrong on 609 lanes, each off by about 2^{32}.

Neither stream is among our results. Under the grader's own input distribution a single shot catches them with probability about 4×10^{-7} and 1×10^{-7}, so its roughly 8,700 shots almost never do.

We close the gap two ways:

1. **A patch to the hhl-angle grader** keeps its random draws and appends a deterministic sweep: every in-domain X is tested once at each of three carry boundaries of k. It rejects both reproducers (469 and 320 wrong-output shots) and still accepts both of the authors' hhl-angle circuits (discovered and the Häner–Roetteler–Svore baseline). It targets this class of error; it does not prove an adder correct.
2. **A standalone C verifier, ucverify,** replays an op stream bit-sliced and checks output, phase, ancilla cleanliness and the Toffoli and qubit counts. For hhl-angle it covers all 2,031,616 inputs X under 45 accumulator patterns, not every (X, k) pair; for coulomb-potential it covers every r_j for four fixed r_i; gaussian it samples.

Under the patched grader, our hhl-angle stream grades OK with 0 failed shots out of 6,103,565 (the receipt is released), and both reproducers FAIL. ucverify also passes our hhl-angle stream on every X under all 45 accumulator patterns, and our coulomb-potential

stream on every r_j for four fixed r_i. The patch changes only the hhl-angle grader, so the other seven tasks grade identically under both graders. Every result in Table 1 therefore holds under both.

7. Comparison with other work

Our best adder has the lowest TQ of any point adder we found that bills every Toffoli on every shot, and is below several that bill more generously. Against record-and-replay and IonQ, whose rules are looser (average billing or more allowed failures), we are lower while paying the stricter bill. The lower numbers we know of all come from ECDSA.Fail, whose scores bill the average Toffoli rather than every Toffoli on every shot: the paper's post-cutoff 1.26×10^9 and a later leaderboard entry. They play by different rules (below).

Point adder	Toffolis	Qubits	TQ	Accounting
Ours, best	999,283	1,317	1.32×10^9	this grader, measured
Ours, published	1,498,991	1,505	2.26×10^9	this grader, measured
Schrottenloher, arXiv:2606.02235	1.9×10^6	1,446	2.7×10^9	cited by the paper
Paper, divstep circuit	4,006,673	935	3.75×10^9	this grader, measured
Luo et al., arXiv:2607.13816	—	—	5.8×10^{10}	the paper's reconstruction, this grader
Record-and-replay, arXiv:2609.28882	—	—	1.92×10^9	reported, looser accounting
IonQ, Häner et al., arXiv:2609.05625	—	—	$\approx1.75\times10^9$	computed from reported 1.196×10^6 Toffolis × 1,462 qubits per windowed addition, lookups excluded; failure bound about 6× ours
ECDSA.Fail paper, arXiv:2609.09582, July best	—	—	1.496×10^9	reported, average billing (26 July cutoff)

Point adder	Toffolis	Qubits	TQ	Accounting
ECDSA.Fail paper, after the cutoff	—	—	1.26×10^9	reported, average billing
Google Quantum AI, arXiv:2603.28846	—	—	—	whole-algorithm totals only (under 70M Toffolis on 1,450 qubits); no per-addition figure, no circuit released
ECDSA.Fail leader, 17 Oct 2026	761,096 (mean)	1,234	0.94×10^9	different rules (below)

The ECDSA.Fail benchmark scores differently in three ways:

- It bills a Toffoli under a classical condition only on shots where it fires, then averages.
- The submitter chooses the test inputs through a nonce.
- It allows zero failures on 9,024 shots.

Its leader reports about 2 failures per 1,000 additions, which would fail this grader's budget of 2 per 10,000. An earlier stream of ours, converted exactly and run through that benchmark's own unmodified evaluator, failed 1 of 9,024 shots there. We therefore claim no record under those rules.

8. Limitations

- **Statistical margins.** The point adder fails 7 shots against a limit of 20, dqi-hermitian 59 and dqi-reed-solomon 8 against limits of 100. A sampled pass is evidence, not proof.
- **Edits, not replacements.** Seven of the eight results are rewrites of the authors' released circuits. Only the point adder is an independent construction.
- **Novelty.** All 28 techniques were checked against a bounded set of primary sources: one has a direct published match and 27 build on published principles. A bounded search is not proof that no other source exists.
- **No security claim.** These are benchmark circuits under one cost model. Nothing here estimates the cost of breaking a real key.
- **Understanding.** The author still does not know what a Hamiltonian is. The grader never asked.

9. Artifacts and acknowledgments

We release each result's op stream with its grade receipt (the grader's JSON), so each result can be re-graded on the unchanged grader; COMMITMENTS.sha256 with the SHA-256 of every result stream (Appendix A); OpenQASM 3 exports (the release README lists which circuits have one; each was checked against its stream's gate counts, not run through a QASM parser); and the C verifier, the two hhl-angle reproducers and the grader patch. The compiler, generators and search tools that produced the streams are not released. Everything released is at github.com/HereLiesRepo/Here-Lies-Space-Time. Our circuits, receipts and this paper are CC BY 4.0, and the verifier is MIT. Seven circuits and both reproducers modify the authors' Apache-2.0 circuits, and the grader patch modifies their Apache-2.0 code; the original parts stay under Apache-2.0, whose text is included, and the patch itself is offered under Apache-2.0.

Credits.

- The point adder's division follows ECDSA.Fail (arXiv:2609.09582) and its dialog formulation from Khattar et al. (arXiv:2510.10967).
- The point-addition adaptation follows Schrottenloher (arXiv:2606.02235).
- The measurement-uncomputed adders follow Gidney (2018). The best adder's squares follow Litinski's controlled add-subtract and schoolbook multiplication (arXiv:2410.00899).
- The other seven results build on the circuits Gu, Kemeny, Yelin and Lukin released with arXiv:2610.22367; we thank them for publishing both the circuits and the grader.

AI assistance. The circuits, verifiers and analyses were produced with extensive help from AI coding agents working under the author's direction: Claude (Anthropic) and Codex (OpenAI). The current hhl-angle and coulomb-potential streams and the prior-art review are Codex's work. Every grade in this paper (Table 1 and every verdict and shot count in the text) is the grader's own output, not an agent report; lane counts and per-shot probabilities come from our verifier, ucverify, and figures for other work are cited from their sources.

Appendix A. Commitments

SHA-256 of each decompressed op stream, as listed in COMMITMENTS.sha256 and in each grade receipt's `h_ops`.

Result	SHA-256
modinv-point-add, best	`394a96b03e0e05c1627340912a1292ed1cfe835b48d04733967d5f6f5e057b42`
modinv-point-add, published	`88be2e5777494671e2227d274ce9cc710348b75e0c7557adc36b8e8c41d9e714`
chemistry-fe2s2	`fe4ba65050aa999d365f793885f5b6af530813b90e5125e2caeff9e116c79501`
dqi-hermitian	`5d58460db308fcce019abe9a1c6336e1391e032b9cf4ab5a33c23986363f6860`
hhl-angle	`2c59af9e93cd947ee7734d731a87039cc9a12561f0e122f071531a9508455985`
coulomb-potential	`35295aa5d6aaa9861cae195959b8baebf1f9b40ee26c39416fdcace068c3cbe3`
lattice-gauge-plaquette	`291fb64deb732c0a2bb25b7f98d8a03b2ecec4eff3f40b8ffd8eeaac0cbdfc13`
gaussian	`c9ca88774d4849c7b61601ab7629606f86fced2b15a9bed528b5880845582ad1`
dqi-reed-solomon	`15be73a0ab99d4ac4694e2aa9bff31d0ac9be5dfd3abd68854be5e56bd44466b`